

Maximum battery voltage (V_{Bmax}) = 14V
MOSFET rating = $2 \times 14V = 28V$

Step 5. Adding safety factor to the current

The safe operation area of MOSFET is temperature dependent. So, in order to make the system more secure, we may take 50% to 80% higher current rating than the calculated current.

Let us take a safety factor of 50%.

$$I_{S.F.} = 400A + 200A = 600A$$

So, the MOSFET we require should be rated for 600A. Therefore, the suitable MOSFET for our operation should be rated for at least 600A and 28V.

SiC vs GaN vs Si MOSFET

For higher powered applications, the silicon carbide (SiC) and gallium nitride (GaN) MOSFETs are replacing the silicon MOSFET. Both SiC and GaN MOSFETs are capable of sustaining higher frequencies and higher voltages compared to the silicon MOSFETs. Higher switching frequency corresponds to lower passive components.

Both SiC and GaN MOSFETs have higher bandgap compared to their silicon counterparts. Furthermore, SiC and GaN MOSFETs have better thermal conductivity and higher electron conductivity, which makes them more suitable for higher power operations. Compared to the silicon MOSFET, the other two devices eliminate tail current during switching, thus reducing switching loss and providing a faster operation.

The table on previous page compares the Si, GaN, and SiC MOSFETs.

In power electronics systems, where a motor needs to be driven, employing a pre-driver or a gate driver circuit is essential for improving the efficiency and stability of the system. The gate driver circuit is mostly used with an N-channel power MOSFET in order to deliver high current to the motor. The design of gate driver circuit and selection of driver IC depends on the parameters of motor.

MOSFETs are the main building blocks of power electronics circuits. Power MOSFETs can be found in high-power applications, such as:

- Switch mode power supplies (SMPS)
- Automotive applications
- Residential, commercial, architectural, and street lighting
- DC-DC converters
- Motor control **EFY**

The author Sharad Bhowmick works as a Technology Advisor at EFY. He is passionate about power electronics and energy storage technologies. Through his work, he wants to help achieve the goal of a carbon-neutral world.



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